Bundle Neural Networks for Functional Connectomes:
A Controlled Test of Density-Dependent Aggregation and Representation Collapse

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Abstract

Graph neural networks appear naturally suited to functional connectomes, but recent controlled evidence suggests that neighborhood aggregation can degrade prediction when node features already encode global connectivity profiles. This study tested whether learned orthogonal bundle transport changes that density-dependent behavior. Using 754 technically eligible ABIDE-I participants from 18 sites, we compared a shared-backbone identity network, a learned-local capacity control, GCN aggregation, trivial-bundle diffusion, and learned BuNN transport across retained positive edge densities of 0, 1, 5, 10, and 20 percent. Evaluation used nested held-out-site validation, five final seeds, regularized non-graph baselines, and common-frame representation diagnostics. The primary BuNN-minus-GCN performance-curve contrast was -0.00958 (95% site-bootstrap CI [-0.03665, 0.01146]; exact sign-flip $p = 0.524$). BuNN was also below the connectome elastic-net reference by -0.05516 (95% CI [-0.08297, -0.02752]). The matched-anchor effective-rank contrast was -0.00719 (95% CI [-0.01310, -0.00105]), opposite the proposed preservation advantage. The small BuNN-GCN contrast was sensitive to site weighting and training seed. Under this specified ABIDE-I pipeline, learned bundle transport therefore showed no detected predictive or representation-preservation advantage. The result is a conditional computational audit, not evidence about biological bundle geometry or universal BuNN performance.

1 Introduction

Resting-state functional connectomes summarize statistical associations among regional blood-oxygen-level-dependent time series. Representing brain regions as nodes and functional associations as edges makes graph learning an intuitive modeling choice. Yet the common node feature in connectome studies is itself an entire row of the connectivity matrix, so each node begins with a global connectivity profile rather than purely local information. In this setting, neighborhood aggregation may remove useful distinctions instead of adding missing context.

Recent multi-dataset benchmarking reported that classical models and non-graph neural networks can match or exceed graph deep-learning models for functional-connectome prediction, and isolated message aggregation as a potential source of degradation [Han et al., 2026]. That result changes the relevant question. The value of a graph operator should not be presumed from the fact that the input can be drawn as a graph; it should be demonstrated against identity propagation and strong regularized baselines under the same data splits.

BuNNs diffuse features after transporting them through learned orthogonal maps between node-specific frames [Bamberger et al., 2025]. Their theoretical and synthetic motivation includes greater control over representation collapse than ordinary graph diffusion. Related neural-sheaf work likewise shows that non-trivial transport can change diffusion behavior [Bodnar et al., 2022]. Whether that advantage transfers to dense functional connectomes with global connectivity-profile features is an empirical question.

The initial formulation of this project connected negative functional correlations with heterophilic message passing. Formalizing the experiment showed that this interpretation was not identifiable: correlation sign does not establish node-label heterophily, excitation, inhibition, anatomical connectivity, or causal information flow, and sign can depend on preprocessing choices such as global-signal regression [Murphy and Fox, 2017]. The dataset, BuNN architecture, connectome representation, and aggregation problem were therefore retained, while the hypothesis was narrowed to a controlled operator comparison.

This study asks whether learned bundle transport produces a more favorable held-out-site performance-density curve than GCN aggregation, whether any advantage exceeds identity and learned-local controls as well as a regularized connectome baseline, and whether predictive behavior co-occurs with common-frame

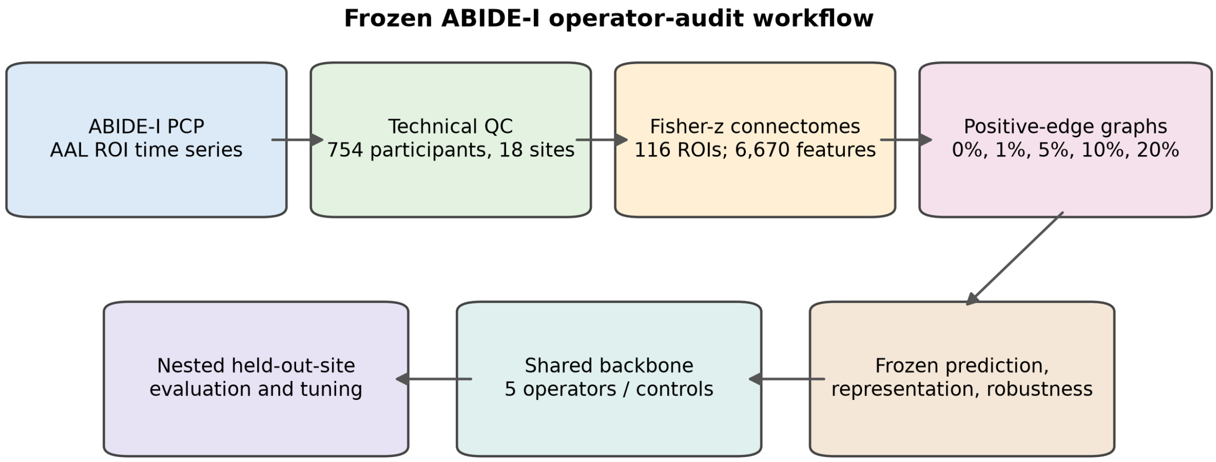

Figure 1: Frozen study workflow. All feature preprocessing, model selection, and hyperparameter selection were confined to training sites. The final held-out site was used only for evaluation.

representation preservation. The contribution is an audit of an architectural inductive bias under one pre-specified ABIDE-I pipeline, not a claim about biological geometry or clinical diagnosis.

2 Related work

Functional-connectome prediction. ABIDE-I aggregated resting-state fMRI and phenotypic data across international sites to support large-scale autism research [Di Martino et al., 2014]. The Preprocessed Connectomes Project subsequently released derivatives from several pipelines, including C-PAC [Preprocessed Connectomes Project, 2026]. Multi-site prediction is challenging because acquisition, motion, demographics, and preprocessing can vary across sites. Held-out-site evaluation therefore addresses a stricter generalization problem than random participant splitting.

Graph aggregation. GCNs apply normalized neighborhood aggregation followed by learned feature transformation [Kipf and Welling, 2017]. Repeated diffusion can contract node representations, although the practical effect depends on architecture, features, graph construction, and task. In functional connectomes, the problem is especially relevant because connectivity-row node features already contain whole-graph information. The benchmark of Han et al. [2026] motivates testing propagation against a no-propagation control rather than comparing only different GNN families.

Learned transport. Neural sheaf diffusion assigns vector spaces and learned maps to graph incidences, allowing non-trivial transport before aggregation [Bodnar et al., 2022]. BuNN specializes this idea to flat vector bundles with orthogonal node maps and diffusion-based propagation [Bamberger et al., 2025]. Raw embeddings from different node frames cannot be compared naively; meaningful collapse diagnostics require transport to a common frame or a gauge-invariant measure. The present experiment therefore separates raw map-generator capacity from inter-node transport using a learned-local control.

3 Methods

3.1 Study design

The protocol was organized as a controlled operator audit (figure 1). Classical baselines were completed and independently audited first. A shared neural backbone was then paired with five propagation or capacity-control variants. All neural configurations used the same outer sites, inner grouped folds, candidate budget, stopping rule, and final seeds. The complete neural artifact was audited without revealing performance values; confirmatory analysis code and decision rules were frozen before unblinding.

Table 1: Frozen cohort and connectome dimensions after technical quality control.

Item	Value
Downloaded AAL time-series files	769
Technical exclusions	15
Retained participants	754
ASD / control	371 / 383
Held-out sites	18
ROIs / lower-triangle features	116 / 6,670

3.2 Dataset and technical quality control

We used ABIDE-I C-PAC derivatives distributed by the Preprocessed Connectomes Project [Di Martino et al., 2014, Preprocessed Connectomes Project, 2026]. The frozen source was the band-pass-filtered, no-global-signal-regression AAL ROI time-series derivative. The AAL atlas defines macroscopic anatomical regions in standardized space [Tzourio-Mazoyer et al., 2002]. A downloaded file was technically eligible only if it had a valid header, finite numeric values, non-zero variance for every ROI, and produced a finite connectome. Of 769 downloaded files, 15 failed the zero-variance requirement, leaving 754 participants (371 ASD and 383 controls) across 18 evaluable sites. These rules establish technical eligibility, not population representativeness.

3.3 Connectomes, node features, and graphs

For each participant, Pearson correlations were computed between all pairs of 116 ROI time series and transformed with Fisher's r-to-z transformation. Diagonal entries were set to zero. The resulting matrix was finite and symmetric, and its 6,670 lower-triangle values formed the classical feature vector. Each neural node received its full connectivity row as a 116-dimensional feature, so node inputs were globally informed.

For nonzero graph conditions, the strongest positive off-diagonal associations were retained symmetrically at densities of 1, 5, 10, and 20 percent. Density zero contained only the matched identity or learned-local anchor. Negative associations were excluded as a pre-specified computational choice and were not interpreted biologically.

3.4 Classical baselines

Three logistic baselines were fit: covariates only, vectorized connectomes with elastic-net regularization, and connectomes plus covariates. Elastic-net regularization combines ℓ_1 and ℓ_2 penalties and is appropriate for correlated high-dimensional predictors [Zou and Hastie, 2005]. Covariates were age, sex, mean framewise displacement, and usable scan length. Scaling, feature processing, and tuning were fit using training sites only.

3.5 Shared neural backbone and operators

The encoder, hidden width, depth, nonlinearities, normalization, dropout, pooling, classifier, optimizer, training schedule, and tuning budget were held constant. The variants were:

1. **Identity**: no inter-node propagation.
2. **Learned-local**: the BuNN map generator and node-wise updates, but no inter-node diffusion, controlling for additional learned capacity.
3. **GCN**: normalized neighborhood aggregation [Kipf and Welling, 2017].
4. **Trivial bundle**: the same diffusion construction with identity transport maps.
5. **Learned BuNN**: learned orthogonal node maps followed by bundle diffusion [Bamberger et al., 2025].

3.6 Nested held-out-site evaluation

Each of the 18 sites served once as the outer test site. Within the remaining sites, four grouped inner folds selected among four equal-budget AdamW candidates using two tuning seeds. The selected configuration was refit with five fixed final seeds. Predictions from a test site were never used for scaling, tuning, early stopping, or configuration selection. The primary predictive endpoint was equal-site mean balanced accuracy; pooled balanced accuracy and AUROC were secondary summaries.

Table 2: Classical held-out-site baselines. Equal-site balanced accuracy is the primary summary.

Model	Equal-site BA	Pooled BA	Pooled AUROC
Covariates-only logistic regression	0.5652	0.5629	0.5973
Connectome elastic net	0.6401	0.6495	0.6794
Connectome + covariates elastic net	0.6336	0.6428	0.6773

3.7 Density curves and representation diagnostics

The primary predictive estimand was the normalized trapezoidal area under the balanced-accuracy curve across the complete density grid. GCN and trivial-bundle curves used identity at density zero; the learned-BuNN curve used the parameter-matched learned-local anchor. This prevented a favorable density from replacing the pre-specified curve-level comparison.

After each propagation layer, embeddings were mapped to a common learned frame before cross-node comparison. The primary representation endpoint was normalized effective rank at layer 2. Normalized dispersion, mean pairwise cosine similarity, invariant edge-transport distance, and encoder-to-layer-2 rank change were secondary. Predictive and representation behavior were tested as co-occurring outcomes; no mediation or causal claim was made.

3.8 Statistical analysis and decision rule

Five final seeds were averaged within each site and configuration before site-level inference. Paired percentile intervals used 10,000 bootstrap resamples of the 18 held-out sites. Exact two-sided paired sign-flip tests accompanied curve contrasts. Density-specific tests were secondary and Holm-corrected within each named contrast family.

A complete positive BuNN result required all three pre-specified conditions: better representation preservation than GCN, better held-out prediction than GCN and no-propagation controls, and performance exceeding the strongest regularized baseline. A confidence interval containing zero was reported as no advantage detected. Absence of a practically meaningful gain was asserted only when the upper bound excluded the pre-specified +0.03 balanced- accuracy margin.

3.9 Run integrity and reproducibility

Long runs used atomic per-site checkpoints, immutable data/configuration/source hashes, deterministic site ownership, resume validation, and independent completion audits. The accepted neural run contained all outer sites and passed the frozen coverage, recomputation, diagnostic, runtime, and warning checks before the result embargo was lifted. Paper values are generated from a hash-bound result snapshot rather than typed independently.

4 Results

4.1 Cohort and classical baselines

The final technical cohort contained 754 participants across 18 held-out sites (table 1). The connectome elastic net was the strongest classical reference by equal-site balanced accuracy (0.6401), compared with 0.6336 for connectomes plus covariates and 0.5652 for covariates alone (table 2). These are cross-site computational results, not estimates of clinical diagnostic performance.

4.2 Predictive density curves

All three diffusion curves declined descriptively as density increased (figure 2). The primary normalized curve-area difference for learned BuNN minus GCN was -0.00958 (95% site-bootstrap CI [-0.03665, 0.01146]; exact $p = 0.524$). The interval crossed zero, so no predictive advantage was detected. Its upper bound was below the pre-specified +0.03 margin, excluding an improvement of that size under this estimand and pipeline, but not establishing universal equivalence or excluding smaller gains.

The learned-BuNN curve was also below the site-matched connectome elastic net by -0.05516 (95% CI [-0.08297, -0.02752]; exact $p = 0.0018$). Neither the identity comparison nor the learned-local capacity control

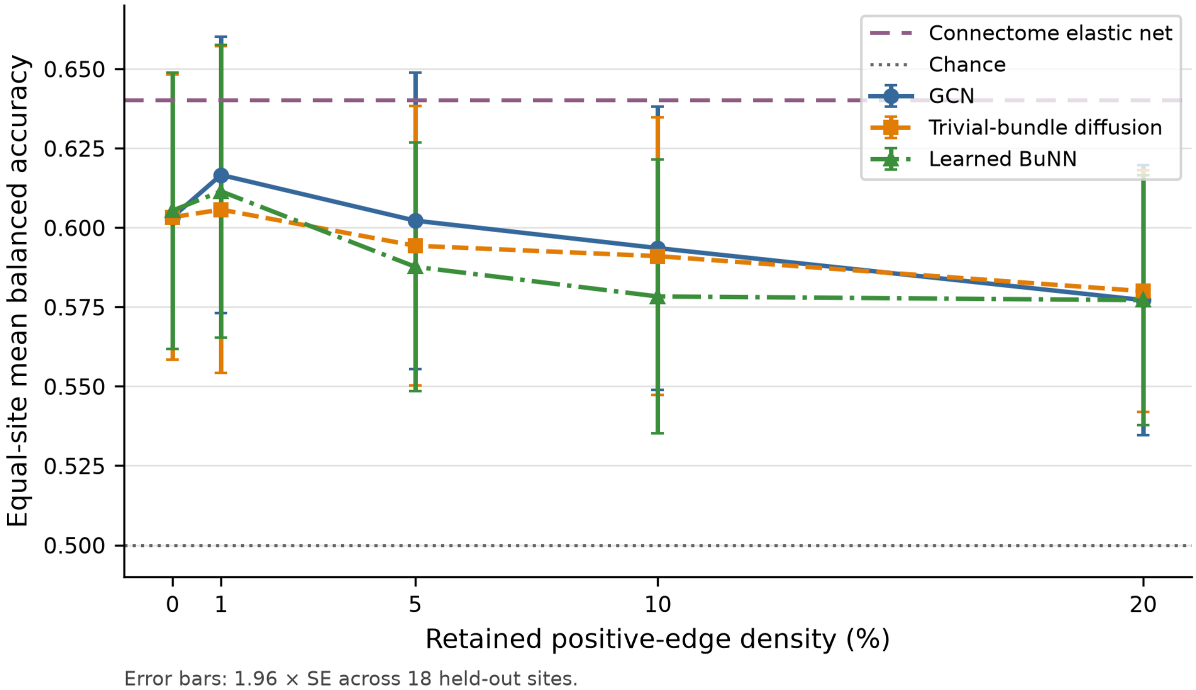

Figure 2: Equal-site mean held-out balanced accuracy across the pre-specified positive-edge density grid. Error bars show 1.96 standard errors across sites for visualization; confirmatory inference used paired site bootstrap and exact sign-flip tests on the frozen curve estimand.

Table 3: Frozen predictive contrasts. Differences are BuNN minus the named comparator.

Contrast	Estimate	95% site-bootstrap CI	Exact p
BuNN curve - GCN curve	-0.0096	[-0.0366, 0.0115]	0.5237
BuNN curve - trivial-bundle curve	-0.0062	[-0.0293, 0.0108]	0.6813
BuNN nonzero mean - learned-local	-0.0167	[-0.0347, 0.0009]	0.0958
BuNN nonzero mean - identity	-0.0146	[-0.0320, 0.0018]	0.1155
BuNN curve - connectome elastic net	-0.0552	[-0.0830, -0.0275]	0.0018

supported the complete positive-result decision rule (table 3).

4.3 Common-frame representation diagnostics

The primary matched-anchor normalized effective-rank contrast was -0.00719 (95% CI [-0.01310, -0.00105]), opposite the proposed BuNN preservation advantage. The raw BuNN rank curve was higher, but the matched learned-local anchor already contained additional map-generator capacity. The controlled contrast therefore did not support attributing raw rank differences to inter-node bundle transport. The secondary dispersion and cosine diagnostics were directionally consistent with the same interpretation (figure 3); they do not establish a biological mechanism.

4.4 Site, seed, and weighting sensitivity

Only 1 of 18 leave-one-site estimates was positive, and no favorable exclusion interval lay above zero (Supplementary Figure S1). Across the five final seeds, 2 point estimates favored BuNN, but 0 favorable intervals excluded zero; 1 unfavorable seed interval lay below zero (Supplementary Figure S2). Participant weighting changed the small BuNN-GCN point-estimate sign, whereas the equal-site mean and median were negative. Equal-site weighting remained the confirmatory estimand.

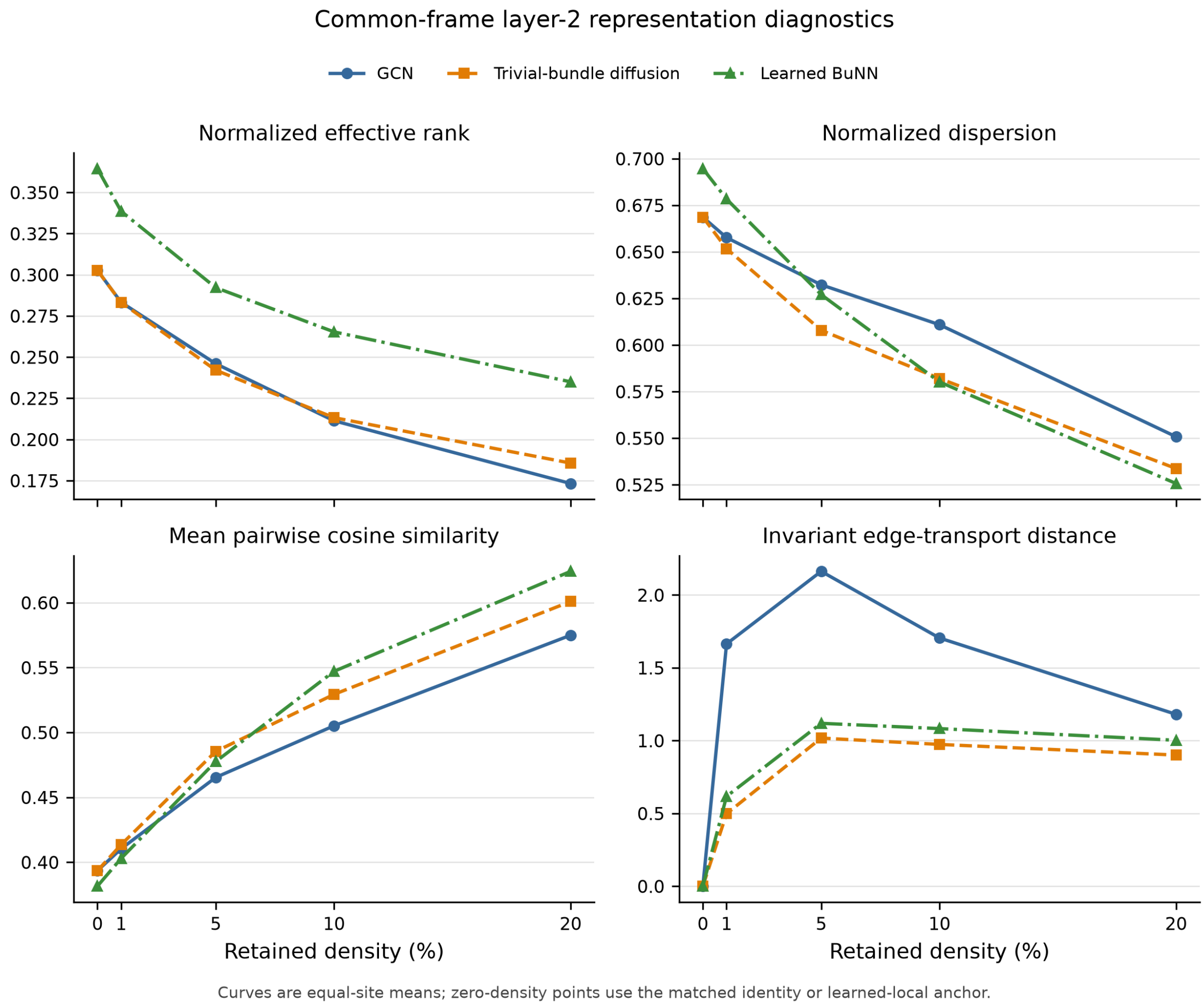

Figure 3: Equal-site means for common-frame layer-2 representation diagnostics. Zero-density points use the matched identity or learned-local anchor. Raw unaligned BuNN node embeddings were not compared.

4.5 Observed execution cost

Learned BuNN used 8,529 parameters compared with 5,889 for GCN. Across the recorded fits, learned BuNN required 31.14 total fit-hours versus 15.09 for GCN, while its equal-site curve balanced accuracy was lower (0.5849 versus 0.5945). These timings describe this implementation and A10G execution, not universal computational complexity.

Table 4: Observed execution cost and predictive curve summary for the three diffusion operators. Runtime is implementation- and hardware-specific.

Operator	Params.	Fit h	s/fit	Peak GiB	Curve BA
GCN	5,889	15.09	20.39	0.373	0.5945
Trivial-bundle diffusion	5,889	25.99	35.12	0.377	0.5911
Learned BuNN	8,529	31.14	42.08	0.378	0.5849

5 Discussion

This controlled audit found no complete transfer of the proposed BuNN anti-collapse advantage to the specified ABIDE-I functional-connectome pipeline. Learned bundle transport did not improve the primary performance- density curve relative to GCN, did not exceed the strongest regularized connectome baseline, and did not show the proposed matched-anchor effective- rank advantage. The result is scientifically useful because the design isolated propagation from backbone, input, tuning-budget, and capacity differences.

5.1 Why transport may not help in this setting

Two properties distinguish this task from the settings that motivate bundle transport. First, functional-connectome graphs are dense even after thresholding. Second, the feature attached to each node is its full connectivity row, so each node already contains global information before propagation. Diffusion may therefore average globally informative profiles rather than deliver missing long-range context. The downward descriptive density curves are consistent with density-dependent aggregation degradation, but do not alone prove a general over-smoothing mechanism.

5.2 Importance of the learned-local control

BuNN contains additional learned map-generator parameters. A comparison with GCN alone could confuse transport with added node-wise capacity. The learned-local control retained this capacity while removing inter-node diffusion. Its high raw effective rank showed why unaligned or unmatched rank curves could falsely appear to support bundle transport. The matched-anchor analysis instead tested what changed when inter-node transport was added.

5.3 Heterogeneity and uncertainty

The primary BuNN-GCN interval crossed zero, and its point estimate changed sign under participant weighting. Site exclusions and final seeds also changed the magnitude and sometimes the direction of the small contrast. These analyses do not reveal a hidden positive result; they show that a small average effect is unstable relative to multi-site and optimization variation. In contrast, the negative BuNN-versus-elastic-net result and matched-anchor effective-rank finding were stable across all leave-one-site exclusions.

5.4 Limitations

The conclusions are conditional on ABIDE-I, C-PAC filtered data without global- signal regression, the AAL-116 atlas, Pearson/Fisher-z connectomes, positive- edge thresholding, connectivity-profile node features, the fixed density grid, the shared backbone, and the equal tuning budget. Functional correlations do not identify anatomy or causal information flow. The study has no locked external validation cohort, and site sample sizes vary substantially. Runtime comparisons depend on this implementation and hardware. Finally, the representation diagnostics measure computational behavior; they do not show that a representation change caused a predictive change.

5.5 Future work

The next scientifically clean extension would freeze the complete ABIDE-I code and evaluate it on a preprocessing-compatible ABIDE-II cohort. Separate pre-registered sensitivity studies could test global-signal regression, negative-edge handling, another atlas, or node features that do not already encode the full connectome. Such studies should remain distinct from the present confirmatory result and should not search across datasets for a favorable diagnosis.

6 Conclusion

Under the pre-specified ABIDE-I pipeline, learned orthogonal bundle transport did not produce a more favorable held-out-site performance-density curve than ordinary GCN aggregation, did not surpass identity or learned-local controls, and remained below a regularized connectome baseline. Gauge-aware matched- anchor diagnostics likewise did not support the proposed representation- preservation advantage. The evidence therefore supports a narrow conclusion: bundle transport changed the operator and increased capacity and runtime, but no predictive benefit was detected in this dense, globally informed functional-connectome setting.

Data and code availability

The analysis uses the public ABIDE-I Preprocessed Connectomes Project derivatives. The reproducibility package records the exact data manifest, technical exclusions, site splits, configurations, seeds, source hashes, analysis scripts, and aggregate result artifacts. Participant-linked audit artifacts remain separated from the public-safe release package.

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